

Active Inference Livestream #002.1 "Is the free-energy principle a formal th

Livestream | 1:26:16

hello everyone

welcome to team com podcast number two

this is going to be a journal discussion

of ramstead at all 2020

and it is going to be a great discussion

i'm really looking forward to it

so before we start off the discussion of

the paper

could the other two panelists introduce

themselves

maybe sasha first sure

hi my name is sasha i'm a neuroscientist

and

a free energy enthusiast

andy vaughn hello my name is iran and

i'm just interested in active inference

stuff

and today we got to

read some soft book

yes absolutely a tough book so

does the screen look okay for sharing my

slides

all right here we go so the structure

of this discussion will be as follows

first we'll talk about some warm-up

questions

then we're going to jump into the paper

discussion we'll

go through the abstract and the big

questions making sure that we really

understand how the authors summarize

their own work

then we'll look at the roadmap of the

paper which are just the section headers

to understand

how they kind of get from a to z
we'll review some of the key claims and
the aims
of the paper so that we can evaluate
later on
whether the claims are justified or
whether they end up achieving the aims
that they set out to
to achieve then we're going to go
through
each of the six figures of the paper
and then um if not addressed during
a through d we'll have extra time at the
end to discuss
any other questions about the paper or
related topics and we have some other
figures that we could put up if we need
to
so here we go with the warm-up questions
first
what is semantics
or what does the word make you think of
when you first hear the word semantics
and you might want to think about
logical and lexical
semantics any thoughts there sasha
um yeah i think this is uh kind of the
getting into the confusing theme of the
paper where you have to be extremely
specific with the language that
you use and so to me semantics means
um how we use words
and how we make meaning
okay very nice and the way that we use
words
the meaning of words is often what
people mean by semantics like syntax is
how it's written

and then semantics uh or at least the semantics of words

is about how the words are related to each other and then also we have logical semantics which is just purely about logical statements and may or may not have any word

representation behind it so there's that word

representation and this is definitely one of the key words of the paper

so maybe yvonne what do you think about or what does a representation mean to you

to me as we can say yeah as

as we as we are saying

we actually represent in our mind different uh

[Music]

different things behind different conceptions

so it's always

hard to understand not always but

usually have to understand what people saying when

when they use one or

another concept yep

and that vagueness about the term representation

is really one of the things that the paper sets out to clarify

because a lot of times people will use the word representation

assuming that one must exist oh if you're thinking of

elephants there must be some neural state in your head that represents an elephant otherwise how could you be

thinking about an elephant um and then
this paper is going to really drill down
what is the relationship between that
conception and the neural representation
um next question what is intentionality
who or what has it so just kind of off
the cuff because of course this is a big
topic but what does intentionality mean
or who gets to have intentionality
either of you
i think intention

[Music]

is
became when
when someone wanted wanted to do
something
yes intention it about the future so
i will do it in the future
any thoughts there um
yeah another wonderfully vague word and
um we often talk about
uh in interpreting other people's
actions
whether or not we can infer their intent
and
of course that gets quite messy but
with how vague the word is you could say
that everything has intention
because it um
you know even a rock knows that it will
continue to stay a rock
in its uh sort of thermodynamic
landscape
very nice um and this is sort of that
pan psychist
twist which is do physical objects have
intention or what about criminal intent
do humans have intention or

how would we even know and then
how are narratives related to semantics
and then i'll just throw on the last one
there are narratives real
or in what sense are narratives real
any thoughts yvonne or we can move on
it's it's
really it's really hard philosophical
questions about are narratives real or
not
agreed i mean this is really and it's
why it's the last question before we
jump into the paper because
narratives are real like harry potter is
real it's really
a narrative but then of course you go
into it a little bit deeper and if
everybody has their own interpretation
of the narrative
well it's real but everyone has a
different reality
and then is that real well that's kind
of what we want to explore
and yes it's a different tool than
people expect
the usual tools that people would expect
to discuss these topics would be
very qualitative and wordy philosophy
arguments
and this is going to be a pretty totally
different approach
involving the free energy principle so
here we go
the paper that we're discussing today is
a recently accepted paper
from the journal of entropy but we're
reading off of the version that we found
on researchgate

and the paper is called is the free energy principle a formal theory of semantics

from variational density dynamics to neural and phenotypic representations

so a lot of new words and for some people this is going to wrangle them because they think it's a philosophical word being used improperly

other people they might not have clarity on what all the words mean

so what we're going to do is look at the big questions of the paper which are thankfully the first two sentences of the abstract and then we're going to run through the rest of the abstract before we go on to the rest of the paper

so kind of just like you're reading this at home this is how we're going to unpack the paper from the beginning to the end

so there's two big questions and again these are the first two sentences of the abstract

their first question or their first uh big goal

is to assess whether the construct of neural representations plays an explanatory role under the variational free energy principle and its corollary process theory active inference

so there's a lot happening here i would kind of jump into the middle

and it's where you start with a free energy principle as a principle meaning it's a paradigmatic it's axiomatic it's foundational

and then there's a corollary which means
like a subsidiary or a secondary theory
and the secondary theory happens to be a
process theory
which means it's not a state theory it's
not about how things are it's about how
they relate
so another example of a process theory
would be evolution by natural selection
it doesn't tell you what state the
system is going to be in but it tells
you
if certain characteristics are present
in the system like replicators with
differential success and heritability
you're going to end up with this process
playing out so the process is active
inference and the paradigm is the free
energy principle
and then the question that they're going
to work on
is to assess whether the construct of
neural representations
so neural states whether those
have an explanatory role or are going to
enrich us
when we're thinking about free energy
principle active inference
and their second question that's tied to
the first is
if there is an explanatory role for
neural representations under active
inference and the free energy principle
which
we think that there will be then they're
going to assess
which philosophical stances in
relationship to ontological and

epistemological statuses of
representations is most appropriate
so they're going to use active inference
free energy principle
to explore how neural representations
are related to the real world
and then they're going to try to
translate that back to the classical
vocabulary and some of the classical
theories from philosophy
okay so let's continue going through the
abstract and this is definitely where
we're going to want to slow down and
make sure that we understand
what each of these terms mean from a
philosophical perspective what are the
baggage what are the assumptions what
are the implications
so they're going to focus on non-realist
which are deflationary and fictionalist
instrumentalist approaches
and they define it in the next sentence
a little bit so we don't need to dwell
there
we consider a deflationary account of
mental representation
according to which the explanatory
relevant contents of neural
representations are mathematical rather
than cognitive
so that's one type of account of
representation they're gonna pursue
the second type of representation that
they're gonna pursue
is this fictionalist or instrumentalist
account
in this view or this account the
representations are scientifically

useful fictions that
serve explanatory and other aims so
looking at this deflationary account in
the middle
and then the fictionalist
instrumentalist account at the bottom
what does that make either of you two
think about or how do
those two ideas differentiate themselves
or what would be another way that you
would restate either the middle one or
the bottom one
um i kind of struggle with
really specifically understanding what
the deflationary account
means and how it differs because
i kind of imagine that something could
be a mathematical representation as well
as a useful story
but
i think this is a common theme that
we'll be coming back to
um for this whole paper is that
um using active active inference um as a
framework it's not just a hypothesis
it's a lens that you apply um
to how you view things and so um viewing
things through
the active inference lens it seems like
these can all coexist in a way that's
not
exclusionary okay very nice yvonne
let me pass now okay i think that you
really
um said it well sasha which is that
um from a different lens the non-free
energy principle lens
these two viewpoints might be uh very

opposed to each other i'm just thinking
at some you know psychology conference
first someone says i propose a
deflationary account according to which
dot dot and the next person says
i propose a fictionalist account dot dot
dot but free energy is this new
lens and this new way of trying to
approach these classical
philosophical problems and the goal
would be that we could get
the best parts of both of them and use
them under the free energy principle
so from the deflationary account we
could
make mathematical representations of
neural
states that would be awesome and then
using a fictionalist or instrumentalist
accounts
maybe we could also have those
mathematical states be useful
so we kind of have like this this triad
we have neural states
i mean the neurons are definitely doing
something we don't know what but they're
doing something
then we have mathematical uh inference
which
what whatever the neurons are doing you
can always mathematically represent it
and then there's utility for the
scientist and or for the organism itself
and it's like you can choose one two or
three of them
and the goal would be we'd want all of
them to be under a common framework
so that's really where they're going

with this approach is instead of just
butting heads
oh well this part of the triangle is
better than this part of the triangle
they're going to see if free energy can
integrate across these different
perspectives
so the abstract's a little long here's
the second part
after reviewing the free energy
principle and active inference
we argue that a model of adaptive
phenotypes under the free energy
principle
can be used to furnish a formal
semantics enabling us to assign semantic
content to specific
phenotypic states which are the internal
states of a markovian system that exists
far from equilibrium so phenotype is
being used here quite broadly
a lot of times people think about
phenotype as measurable components of an
organism
and that's like oh well leg length is a
phenotype or hair
color is a phenotype and they are but
also
the instantaneous representation of the
neural
uh system is like a phenotype and so
just like a phenotype could be
reaction time another type of phenotype
could be
when i play a 200 hertz hone tone
into your ear your brain state in
response to that tone
is a phenotype so here phenotype is just

meaning

measurable components of a biological system whether you actually measured them or not

and then um they're going to ask

whether they can assign semantic content so

that's meaningful content either lexical semantics which is how words are related or logical semantics which is how ideas and propositions are related and they're going to ask

whether they can make some sort of a mapping between

semantics and phenotypic states within a markovian system

and by markovian system far from equilibrium they're highlighting

two pieces to call the system markovium means that it can be bounded within a markov blanket

which is to say that there are states that are internal to the system

and states that are external to the system and then the boundary nodes

um where when you're inside looking out it's what you see or outside looking in

is what you see and that's this markov blanket idea that we kind of come back to in a lot of our discussions

they're going to propose and this is why

i made this font big a modified

fictionalist account in

organism-centered fictionalism or

instrumentalism so they're going to say

um

let's be pragmatic let's make a modified account

where the organism is in its own
generative model
making an instrumental fiction about the
world
where uh i'm not exactly sure all the
connotations of instrumentalism here
but it's kind of like the body is an
instrument and then the song that's
being played
is the niche um is the fitness of the
organism's phenotype
to the challenges of its niche
they argue that under the free energy
principle pursuing even a deflationary
account
of the content of neural representations
licenses the appeal
to the kind of semantic content involved
in the aboutness or intentionality of
cognitive systems
our position is thus coherent with but
rests on distinct assumptions from
the realist position and so here we get
another
peek at the difference potentially
between the deflationary accounts and
the inflationary accounts
which is not really used but what does
it mean to deflate
or inflate in this context deflationary
it's like if somebody said oh it's just
a party it's a deflationary account of
the party it's like it's nothing big
okay whereas the inflationary account of
the party
would be like this is going to be the
most important party ever
it's going to be so important that all

these things happen in this really
symbolic way
and so the deflationary account of
neuroscience is it's
just what the brain is doing it's just
cells that are linked up to each other
and it's just time is passing and click
by click
in time it's just the physical system
but then you get into a little bit of a
bind when you want to reintroduce
semantics because if you just have this
massive agent-based model or dynamical
model
that's just being simulated forward
through time then
you can say um well it's as if it's
thinking about
elephants but you're never going to be
able to go back to the micro state and
say this is elephant
in the microstate as it's represented
because it's deflationary you kind of
deflated the meaning away
and um that's the realist position um
whereas this semantic is an inflationary
account it's like yes
it's a brain state but it actually is
meaning something more
and so that's what they're going to try
to take the best of both worlds they
want to have a formal
semantic system um like philosophy does
formal semantics but they want to ground
this formal semantics in how it arises
from phenotypic representation
specifically neural states
and then this is their closing argument

that we argue that the free energy principle
thereby explains the aboutness or intentionality in living systems
and hence their capacity to parse their sensory streams using an ontology or set of semantic factors and so here is where they kind of
uh reach their synthesis and their proposal
which is that if we can find some sort of a bridge
between realism like what the brain is actually doing
without any extra fluff just phenotype and a formal semantic theory potentially this will light the path forward for understanding how
organisms parse their sensory data coming in
into meaningful categories so instead of just saying
well the photons hit my retina and within a predictive processing framework my priors are updated and
yep there's sasha in the window and there's evon in the window instead of that type of
processing forward it's actually like we get to think about
philosophically what the organism is doing through parsing its sensory stream but also retain the realism of the sensory stream okay
any thoughts on that relatively long abstract
it's more clear now okay that's the only hope

if we aren't i would say the same good
if we're not reducing our uncertainty
about this paper
and their claims and their aims and
their goals then we're not really in it
so now that we have a little bit of a
better sense of
where they want to come from and where
they want to go to
let's look at the roadmap this is going
to be the path that we take
from the introduction of the problem and
its relevance
through the foundations to the claims
that are synthetic
so they're going to begin in the
introduction by talking about the idea
of neural representation
and the contents of neural
representation as well as a little bit
of a joke the dis
contents like dis discontented means to
be unhappy
so there's some issues with neural
representations um
and they're going to be discontent with
them by talking about the problems with
their contents
so classical joke the faces of
representationalism
realism and non-realism and so here is
those two different
dueling perspectives on representing and
it's another joke it's two faces
two faces of the same coin so if you're
gonna have representation
um it's kind of sitting just like a coin
on its edge

on one face it's the semantics what does
the

representation mean you're thinking of
an elephant it means elephant
and then there's the deflationary oh
it's just a neural state
and there's actually no meaning attached
to it even though it is a representation
of an elephant

then they're going to take a turn
towards anti-realism

they're going to talk about some of the
deficiencies of the realist view
and there's a lot of sort of ways to
approach this we'll look at the
specifics

but one example that i always come back
to is like

most people don't perceive their blind
spot or

they perceive that they have color
vision in their whole field of vision
and so yes the world really does have
wavelengths of different colors
coming from all over our visual field
but that's not

really what you're perceiving in your
retina so it

really is being generated by your brain
so it's really an organism's centered
fiction

which is why they're going to come back
to that idea later then they're going to
address

how representations are discussed
under the free energy principle as a
paradigm the second section

is where they turn more heartily to the

free energy principle
and they talk about active inference
again as a corollary process theory
to the free energy principle as a
paradigm and then specifically
they're going to bridge two topics here
they're going to move
from information geometry to the physics
of phenotypes
those might be two very new terms or two
way
new ways of thinking about information
and geometry a lot of times people think
about information theory is one type of
science and then geometry you know
triangles and squares
it's different so what is information
geometry
and then how is it related to not just
phenotype but the physics of phenotype
so first they go through state spaces
non-equilibrium dynamics and bears
oh my again hilarious writing
they're going to talk about markov
blankets and the dynamics of living
systems
so again a markov blanket is like this
set of nodes you can think of around a
system
that is the ideal perfect clean cut
carving up nature at the joint between
the internal and the external system
and going into the internal system is
sense data
and leaving the internal states are
action states
so we'll look more through that soon
then they're going to return to

information geometry and the physics
of not just phenotype but sentient
systems
and that's going to have to do with that
far from equilibrium nature
of living systems and it's why the rock
is uh not the person the object
is not alive but a person uh who has
far from equilibrium thermodynamic
states they're considered to be
living then they talk about phenotype as
a tale of two
den cities again trying to loop as many
puns as possible into their titles and
then they close out section two
with the discussion of living models
which is a mechanistic view
on goal directed probabilistic inference
and decision making under the free
energy principle
so everything is under the free energy
principle because that's
um our working model and then here
they're going to try to go from a
mechanistic
view so they want to talk about the
mechanisms which are how things are
actually
happening but they also want to include
some things that are not traditionally
included in mechanism
for example goal directedness
probabilistic inference
and even decision making then in section
3
is where they uh bring together their
deflationary and fictionalist account of
neural representation

first they're going to propose their
deflationary account to neural
representation
in the other words brains are just doing
brain stuff nothing more
nothing less that's what fristen calls
deflationary
um in a sense don't don't inflate your
hopes that there's going to be some
magical
secret you know between the neurons in
the brain the brain
is just doing brain stuff and it's
tremendously complex and it involves
multiple cell types
but it's just that nothing more and
nothing less
then they're going to talk about
fictionalism and how models come into
play
in scientific practice and then in the
closing section four
they're going to finally reach their
synthesis which is a variational
semantics from generate generative
models to deflated
semantic content so now they're going to
try to deflate meaning
itself so when i say that my name is
daniel and there's so many layers to
interpret that on
it's a string of audio bits now it's a
string of audio bits that is being
transmitted through the computer
through the live stream it's just
exactly what i mean
it's just the signal i'm sending nothing
more and nothing less

so how are we going to get this kind of
deflationary realism about the world
that's so critical for an unbiased or at
least transparent
scientific approach we want that
unbiasedness
that objectiveness yet because
we're semantic entities we know that
we're going to have to engage with the
world
in terms of meaning so how are we going
to square this circle with
realism and what is just there with
meaning and all the power that meaning
based approaches can bring to the table
then they're going to um yeah provide
the deflationary account of content
under the free energy principle and move
from
a computational theory that is proper to
a formal semantics
and then they're going to close out with
the idea of phenotypic representations
and ontologies
okay big road map a lot of stops a lot
of gas stations a lot of restaurants
any thoughts on that
okay good sasha
all right yeah that was really helpful
um
to talk through the the road map and um
see where we end up okay
so sorry for the wall of text but i
wanted to
put one of their key claims verbatim
they write and this is a
pretty early on in the paper i think
this is on the bottom of page

uh the actual paper
uh page uh three or four yeah bottom of
page three in the
pdf but the page numbers could be
different for a future version
they write there are several well
accepted constraints for the
appropriateness
of representational explanations so
we're talking about
representations and now we want to think
what are the
desirada what do we want what would be a
great theory of representation
and they're going to play by the rules
of people who study representation
they're going to say these are the
community of people who study
representation they've decided that
these are like
the five rubrics if you can ace all five
of these tests
then you're going to get an a plus for
your theory so first it should cohere
broadly
with actual practices that are used in
computational cognitive science
research so that means if you have some
theory of representation that only
applies on a different planet or outside
of an fmri machine
you're probably not going to get too far
we want this representation model
to be grounded with the real things that
people are doing
in computational cognitive science and
also psychiatry
um two we want to allow for

misrepresentation
which means that the representation has
to be able to get it wrong
and so if you say think of an elephant
and then anything that the person thinks
of you're going to say right well that's
what they their version of elephant is
they started thinking about lunch
but they can't be wrong because no
representation is incorrect
that's not going to be super useful we
won't be able to actually parse the
world
into like useful pieces of information
and not useful
um and that's a lot like saying if you
can't um
if you can't have a model that gives you
the wrong answer it can't give you the
right answer you know if it just gives
you a one
no matter what you put into it it's not
useful just because it might be right
twice a day like a broken clock doesn't
mean that it's actually a useful model
three representation should provide the
principled method
for attributing determinate contents to
specific states or structures
internal to the system um so we're
looking for some sort of a bridge
for a way that we can uh go from a
specific brain state
like your amygdala is having a lot of
blood flow and your prefrontal cortex
isn't that would be a very coarse level
representation
or uh you could imagine a a one million

dimensional representation
this neuron is doing that and this glia
is doing that we want to go from
that uh specific state representation
which might look like a vector or some
other map and we want to go from there
to something that is internal and or
external to the system like
thirsty or hungry or thinking of an
elephant
finally four we want this representation
theory to be naturalistic meaning that
the account of semantic content does not
itself appeal to semantic terms
when defining how the representational
capacity
is realized by the physical system on
pain of circularity and reasoning now um
pretty rich for the free energy
principle to call someone else
out on the circularity of reasoning and
it's uh it's a complex claim to parse
what is being said with number four
because
in some deep sense because our um
axioms are chosen without evidence by
definition
our paradigms uh are non-explanatory
they can't be explained
uh the question of what's a circular
reasoning argument and what isn't
um is is not perfectly well defined but
i think what they're getting for uh
getting out with number four
is like we want the semantic content
of the elephant to be an elephant not
just the word elephant
we don't want to enter into this

infinite recursion where like
language explains language and then that
explains another kind of language
and we just stay within this realm of
abstracted uh
representations but we never actually
come back
even in a uh sort of unexpected way
to the reality of the naturalistic
situation
which is as evolved entities we actually
have evolved to represent certain
external patterns
internally and so if it doesn't come
back to that naturalism
then it's just gonna fly off into space
any
thoughts on this uh four claims about
what would make a good representational
theory
um i found that last statement quite uh
iron ironic and circular um because in a
way we're always using semantics to
describe the systems that um
are semantic but also maybe eventually
linked to
a natural state or uh you know a real
elephant
if you will um and i
i think uh yeah this paragraph was the
most helpful in trying to
parse out the kind of um
uh things that they're looking for and
the arguments that they're trying to
make
and um i really i think
part two is quite interesting and then
they do get into it later in the paper

but
um to me that's always the most
interesting part of the free energy
principle to address
cool and here just to put it up on the
screen one more time
this is from page four in the uh
accepted manuscript version where they
uh return to the two-fold aim of the
paper
which is to determine whether or not
neural representations
are going to play an explanatory role
or any role at all within this
increasingly popular
framework for the study of action and
cognition
the variational free energy principle
and then second
they want to postulate whether the
neural representation theory
is gonna be warranted under the
principle so this is related to those uh
first two
things that we put up on the screen the
first two sentences the abstract the big
questions
okay so let's jump into the figures and
then we're gonna
work our way down the paper and sort of
landmark
at the different figures so that um
we're understanding how they're visually
as well as conceptually representing
these different topics along the roadmap
and then at any point if you have a
question
about the figure or the text

specifically

or just a related thought that comes to
mind just bring it up

so the first figure and the caption is
copied below

so you can okay

is a markov blanket and this is very
similar

to other images that we've seen

but this is just a foundational term for
the free energy principle is this idea
of a markov blanket

so we've talked about markov blankets
before maybe would either of you like to
give it a shot at what a markov blanket
might mean or

not just the total definition but even
just one aspect of it that you think is
interesting or relevant to remember
right now

okay let me try once have

its integral internal states and
sensation and action and
also in

in space around there is

external states that one

doesn't know about and doesn't know
how uh

how it will

impact on its its

own internal state yes there is a border
that

[Music]

where external states it's out of the
border and

internal states and the actions and
sensation is

in it perfect so here the purple node

mew

is the uh system of interest this is the

project in focus this is like the

internal state

and then the blanket actually goes one

layer out

from the internal states and it goes in

two different directions this blanket

extends it extends upstream

towards the green s's those are the

sensory states

that are coming in to the internal state

through these hidden external causes the

orange nodes

and then leaving the markov blanket

are these a action states and action

states

um are basically resulting from the

internal states dynamics as well as

other

sensory causes so here that s on the

left side

is just going directly to action so when

i was thinking

what does it look like for the sense to

enter the markov blanket

but not enter the model i was thinking

of like um

you know having uh like a splint on the

arm

it's like the overall outcome action

moving your arm it's going to be related

to your internal states

as well as some external influence that

wasn't really part of you

but it's part of your integrated unit of

sense and action and so that

that that external cause about the world

having a constraint on your arm
ends up changing your action states and
therefore it makes sense to include
its sensory input and action uh
influencing ability within the markov
blanket
even though it's not within your
epithelia and that is actually why
there's such a natural transition
from thinking about this type of markov
blanket framework
to thinking about extended cognition
exocortex
ecological cognition distributed
cognition because it's the realization
that even one person's internal states
for example the brain
are in feedback with external tools like
the computer
and so they're external from the point
of view of the epithelium
but actually from the perspective of the
markov blanket they're internal to the
same
self-uh regulatory cybernetic system
that's grasping
at these external causes and trying to
influence them
so i'm going to return just to the paper
so that we can scroll through
and um see where this figure comes up to
but i just wanted to get that first
markov blanket in
so here we are in this section 1.1
with the realism and non-realism and so
here's where they talk about realism
about neural representation
what is real finally we get to find out

um

this view combines two positions

ontologically

that neural representations really exist

which means that they're physically

something that's instantiated

happening in the brain and then

epistemologically

that's how we come to know about things

that representations are scientifically

useful as well

so realism they're saying has an

ontological component which is that

they're real

they're actually physically instantiated

and epistemological realism

is that scientists should find this

realism

uh useful non-realist positions um

which have not a complete overlap and

that's a little bit of the confusion

is it's not like the realists and the

non-realists are totally opposed

they do share some aspects of their

models so it's not like

perfectly everything that realism is

non-realism isn't

so what what does that mean non-realists

are

either agnostic about the reality of

neural representation

so that's kind of just like saying they

don't care what's real in the brain or

they'll never know or we couldn't

measure it anyway there's a lot of ways

you could be agnostic

right about anything or they explicitly

reject the assumption and so the

non-realist is kind of like the agnostic
the anti-realist is like the strong
atheist
they say neural representations do not
exist
so that's the strong claim is
anti-realism or realism
theism atheism and then non-realists are
kind of like
the utilitarian the agnostic in the
middle
and they continue talking through
several of these different
philosophical ideas or these schools of
thought like
eliminationism which is anti-realist and
it's basically saying
the construct should be eliminated so we
shouldn't even use it
and that reminds me of like the idea of
a genotype phenotype map
or a mapping some people say yes there
is a map and we can find out about it
other people say no there's no map but
it's as if it exists
so you can talk about it it's useful but
it doesn't really
exist and then there's other people who
say you should eliminate the concept of
a genotype phenotype map
because it just is simply misleading to
scientists so that's eliminationism
and then instrumentalism or fictionalism
is non-realism and is saying that
they're useful
so that is hearkening back to the work
of dennett
for example with the intentional stance

as well as a lot of other
folks research um just
speeding through this because there's a
lot of sections they're going to talk
about parallel distributed processing
they're going to talk about some of the
issues of realism
um and we
we're just gonna move through this
because the specific critiques of
realism um
i feel like they're gonna end up moving
beyond them so
if you're curious you can read more into
their specific
problems with realism then they're going
to return to
representations under the free energy
principle and that closes out section
one
so i think we could then go to um figure
two to
look at the free energy principle so
here is where we
get to figure two um
and where figure two is conveying the
basic
outline of active inference in the free
energy principle so would either of you
two
like to take a stab what do you see in
this figure either on the left side or
on the right side
um sure i'll take a stab
yeah so this is uh kind of taking that
model of the marco blanket in figure one
and um trying to uh
fit it to some of the parameters of um

an organism
and you know with the image of a brain
to remind us what we're doing here
um and
to explicitly say and mathematically say
what
external and internal states are
representing
and that they're linked by the sensory
states
and active states now sensory states
are explicitly going into the internal
states
and active states are explicitly going
out into the external states and so
and then there's feedback between um
some of the the components in this
system
very nice uh yvonne any thoughts on this
uh as
as was as was told
at the start that we can
we can have a mathematician
mathematician expression of
of the states and there is such
expressions in math so
if we have
if we stay different well yes we can
describe how
we propose to
have it in math yes
it's also it's also has
w with a little note down
how it's letter called i don't know
omega omega
oh manga okay
and yeah what is it that i just forget
it's it's random noise yeah so that is

kind of like a error term

in each of these uh four so this is

adding another level of complexity to
this figure

so here we had hidden external states
that are orange

sensory states coming in that are green
purple internal states

blue action states going out and then
influence on those latent causes in the
world

in orange at the bottom now on the left
side

we have internal states with blue that's
still the μ

variable and it contains some internal
function

f of μ and b and so b is the blanket
and it's basically saying

um μ is going to be a function of
 μ and b so the state of the internal
state

is a function of itself and the blanket
states

and there's a noise term okay then we go
to the active states

active states are a function of μ and b
they're a function of the internal state
and

the blanket states that makes sense plus
another noise term

and then you can continue working your
way around and you can see that external
states are a function of

external states in the blanket plus a
noise term

and sensory states are a function of
external states

uh and the noise term external states
and the blanket state
plus a noise term so this gives us gives
us
a few nice partitionings first we can
talk about
uh doing inference on the variables that
we care about
by separating them away from a noise
term and then also
you can already see that sensory states
don't rely
on internal states or action states
other than the ones that are in the
blanket and so this type of partitioning
is moving us
towards a very clean way of talking
about
these different factors and how they
influence each other
let's do one more figure before we
return to the paper
so here is figure three this is going to
be
another representation of free energy
principle
active inference so here at the top of
the figure
we have this magical g expected free
energy
um and the expected free energy now
we're like looking inside the brain
uh that's the the inference organ that's
calculating expected free energy which
we'll come back to mathematically what
it is
and the brain or the internal state of
the system

guides policy selection pi for policy
like a p
and policy selection is intermediate
between
the model of the world and the actual
action selection
so for example if i see the ball coming
out in front of me
and my policy is that i'm gonna run and
uh try to catch the ball
it's like halfway between action which
is actually
the legs moving and running to catch the
ball and
the pure internal generative model which
is like
wow if i ran as fast as i could i could
catch that ball
in between is the policy choice and so
that's what the brain is trying to
converge upon
good solutions for and that's why um
free energy principle
is adjacent to cybernetics control
theory
and a lot of other areas that actually
are action oriented
because in the end it's not just that we
go from model
to action it goes model to policy to
realized action
um and it's a nuance but it turns out to
simplify
the the uh the system and the model a
lot
then we have initial
uh prior beliefs about the initial
hidden states are in d

so on the left side of the image and
time is going from left to right here
the left side is this r priors about the
initial
hidden state like um i think it's dark
outside
then um the actual hidden states are
these z_t
these news i guess um and actual hidden
states
are going to move through
a likelihood mapping a and result in
sensory observation
so initially in d i believe it's dark
outside
and let's just imagine that this n_1 is
actually dark
so then there's a mapping where if it's
dark no photons appear
and if it's light there's a bunch of
photons and then that results i don't
know about n
i don't know about a but i do know about
 s
which is i don't get any photons so
that's consistent with it being dark
because there's a mapping between it
being light and photons and it being
dark and no photons
um sometimes it seems like we're
explaining things
like multiple times and saying it the
exact same way
but it turns out that by observing um
this structure here
we can see how these beliefs change
through time so
then b are state transition

probabilities

and so then by observing the lack of
photons in s_1

that is uh confirming my belief that
it's dark outside

and so we continue iterating through
time where

basically the probabilities of state
transitions

are happening about the external world
and then the external world keeps on
emitting

sense data to us through this likelihood
mapping

and then what the internal state is
doing is by minimizing free energy
expected free energy across the whole
thing we're gonna come to adaptive
policy selection

okay any thoughts on this sort of third
representation of

markov blankets slash free energy sash
um yeah uh that was really uh useful to
walk through because

um i really wasn't sure how to uh
which direction to approach this figure
from um

and uh the way you described it
it just makes it very clear and reminds
me a lot of um

kind of experimental design where you go
from

one state uh

one set of uh hidden states and then you
observe something

and then you change one variable and you
observe something else

but it really depends on what your

understanding of the
um kind of relationship between you know
photons and darkness for example that
helps you interpret
your next observation um so
i i think it just highlights that we're
always doing that we're always trying to
link um
the observed state to what's causing it
or
kind of a hypothesis testing all the
time
yep and it this is a discrete model on
the bottom it says um the form of this
generative model
is basically discrete but we could
imagine it let's just flesh out that
experimental metaphor
so let's just say the initial hidden
state is that
a bacteria of interest is present or not
in a sample so if it's present there's
going to be dna corresponding to this
bacteria if it's not there won't be dna
that's the likelihood mapping a and so
initially we don't know
and so we do an experiment we perturb
the system by adding some chemicals by
doing a pcr reaction
and then we get the sensory data output
and it it's uh
inclusive and then we we can update our
policy
until we get a sensory uh observation
that's consistent with um what is
happening inside the test tube
and that's why we have positive and
negative controls because if you just

run one experiment
then and you get a negative result well
on one hand
it might be that there's no bacteria
there but also
the likelihood mapping of uh makes it so
that well there could be a bacteria but
then it maps onto a negative result
because the pcr didn't work and so
that's what comes back to the mind of
the scientist whereas
the scientist has to look in a sense at
the sensory data
at the measurements empirically measured
uh results but actually look
beyond them to reduce uncertainty about
something that's
real like whether or not the bacteria
are in that sample
because it's not just did the test come
back positive or negative
that's going to be misleading because
that makes it so that the
s is the same as the hidden state but
it's not the hidden state
it's the outcome as mapped by a from the
hidden state
given your experimental setup all within
a policy situation or an experimental
research program
that's dictated by minimization of
expected free energy
any thoughts on that yvonne or we'll
jump back to the paper
cool so um
in section two let me just look at what
the next
figure is the next figure okay so

we're gonna a few minutes we'll jump to
the paper so
um this is going to be an introduction
to
state spaces and non-equilibrium
dynamics though i'm not quite sure where
the bears come into play
this is a key sentence living organisms
maintain phenotypic integrity and resist
the tendency towards thermodynamic
equilibrium with their ambient
surroundings
full stop this is this is what it means
to be alive
this is schrodinger's 1944 what is life
he says life is going to be locally
thermodynamically organizational
it's going to have negantropic
characteristics it's going to locally
organize
matter and energy not violating the laws
of physics because in the end the global
system
is disordered um but still the living
system creates local order
that's why you need to eat a few
thousand calories of food
to build one pound of muscle for example
um and usually the fluctuation theorems
that generalize the second law of
thermodynamics
would tend towards dissipation which is
why
uh things tend to break down unless
they're being actively repaired and
rejuvenated like living systems are
living systems maintain their phenotypic
integrity so they keep their leg at the

same length or they keep their brain
alive by bounding the entropy
which is also the dispersion or spread
of their constituent states
and so it's like my arm exists in an
organized state right now
um where the bone is bone and the muscle
is muscle and the fat is fat
but you could imagine that if that were
to break down so that just the carbons
went floating free
then the organism would cease to exist
so organisms exist
by virtue of and by merit of
maintaining their physical phenotypic
representations
coherent if it's not going to be a
coherent phenotypic representation
it's gone to get a better handle on this
they introduced two formal notions the
state space
and non-equilibrium steady states so
there's of course a lot that could be
discussed about state space
but a state space is a formalism that
allows us to describe
time evolution of a system by depicting
a trajectory through state space
so a physical state space could be like
your latitude and longitude
and then the state space trajectory is
your path through
latitude and longitude but also you
could have a state space that has
other axes that aren't latitude and
longitude um
that are for example neural state space
and that's bringing us back to this

question of neural representations
um the probability density that
describes the system
at non-equilibrium steady states aka
their phenotype
states are aptly called non-equilibrium
steady state density
so my arm appears to be at an
equilibrium in the sense that it's not
growing or shrinking
so why is that called non-equilibrium
steady state well it's a steady state
because in one respect the phenotype is
unchanging
but that unchangingness is actually an
active
compromise that's being reached between
the forces of order and disorder and in
that sense the total system
is not in equilibrium the equilibrium
system is the rock
at the bottom of the hill that's the
water you know when it's just at the
bottom of the river
there's nowhere to go from but when you
have these
two counter-regulatory processes of
organization and disorganization
you end up with this very dynamic uh
location in state space that's always
being re-optimized
and the universe gives us the disorder
for free that's easy
it's easy to imagine how things fall
apart but what life succeeds at doing is
maintaining
order even though there is disorder and
then

um this idea that there's a
non-equilibrium steady state density
is uh consistent with far from
equilibrium thermodynamics and
this is just related uh to again to
schrodinger's question which is like
the living systems have to maintain
order
somehow they have to maintain far from
equilibrium states
the equilibrium state is all of your
carbons are unlinked and they're blasted
off as CO_2 into the atmosphere
so that's going to be the most well
mixed thermodynamic state
but we're not that we're phenotypically
coherent so
what are living systems and what allows
them
to stay organized despite that
dissipative force
that is where the markov blanket comes
into play
it turns out that by using the markov
blanket formalism
at first just as a scientific heuristic
but we'll see it's actually deeper than
that
we can make clean distinctions between
organismal internal states
which are doing inference on the
external states of the world
active states which are the active
states that actually move the organism
uh actively into a better realm of state
space
like a better temperature for you to
exist within

um and then so we already talked about this

partitioning but now we're thinking about this in terms of the internal states

being far from equilibrium

any thoughts or questions on that

okay so here's where we're going to get to the information

geometry as well as the physics of sentient systems

so this is a big topic we're not going to have

uh 50 hours to unpack it but this is why

we're part of a journal club series

because these are the ideas that we want to come back to

again and again the description of a system

the in terms of movements in internal phase space

is the system's intrinsic information geometry

closely related to measure theory and statistical thermodynamics

so regular geometry would be in the phase space of the piece of paper

you know x and y coordinates geometry is like a triangle has a certain

shape and then you can imagine there's a different state space like

non-euclidean geometry where the triangle has different rules

but you can still talk about geometry

and maybe even things like triangles

but they don't necessarily need to

correspond to the angles must

sum to 180 because it's about the total

systems representation

about the rules of the system and about
the geometry of the system and the state
space

the system exists within all those are
really well linked

so information geometry is asking
about how we can do geometry not on
x y on the piece of paper not on
latitude longitude with a map
but how could we do geometry on the
informational internal states so
it's a it's a big idea and there's a lot
to it but

what is that um what does that ring with
either of you two

and here's go ahead sasha um it's
yeah it sounds like kind of a different
way to phrase um

like topology or mapping relationship
between different

um ideas or different people

so thinking of like network uh

analysis on these different concepts

yep and here's that physical metaphor

where in euclidean geometry

if you move 100 meters then you'll move
100 meters

so there's a one-to-one relationship
between the physical movement
and the euclidean informational distance
however if you're on a sphere if you
move

in the same direction you're gonna end
up zero and so that's like kind of a
modulus operation in mathematics
and as opposed to saying well 10 plus 10
equals 20.

okay but what if you do modulus 10 then
10

goes to 0 and 20 goes to 0. so those
numbers 0 10 and 20 are different in the
number line in the euclidean world

but there's another system another state
space representation

where actually 0 10 and 20 are
the same point in that state space which
is to say that their informational
distance is zero

another interesting thing about bringing
it to the concept of information

is that the informational divergence
between two things is always positive
so you can have um and that's kind of
like um

the distance between two things is
always going to be a positive number
um you could say um on this number line
you know 10 minus 30 is negative 20 but
if you have something on the 30 yard
line in the 10 yard line of a football
field

the distance between them is positive so
the number line is where we get things
like

negative numbers um but within very well
specified systems

information geometry also can tell us a
lot more than uh

other approaches here i don't think
in this specific conversation we'll have
too much time to go into the
equations of information geometry where
they use the fischer information metric
as

a sufficient statistic that corresponds

to the expected
thermodynamic and in external states and
internal states
um but this is let's put a flag here so
that we know that we can come back to
talk about this
when we have some more colleagues on the
conversation
then they get to phenotype as a tale of
these densities
and they say that the phenotype is a
free energy principle is a story about
two probability densities
the first is this non-equilibrium steady
state density itself
which is the statistical structure of
the phenotype so that's that
realist that's the deflationary realism
and the second
in the sense that this is the phenotype
like your arm is a representation of the
arm
it's the uh it is itself the second is
the variational density
which is parameterized by the internal
states of the system
and so that's where we're gonna already
see that we can reach into that semantic
world i'm gonna just um skip through
this part so that we can get to the the
syntheses of the paper and the last
figures so here's where we get to
figure three so now we're seeing that
through
this pretty elegant and minimalist
structure
when the organism minimizes its expected
free energy

which is always going to be a positive number because the informational divergence between any two distributions is positive

if you minimize g you're going to come to the best π

the best policy selection and that policy selection

is going to use input from the emitted sensory states

to reduce uncertainty about hidden causes in the world

um

and this is where they get to

the questions about content of representation

and so here is where they're going to return to those

uh the desired features of neural representation

but here is where they approach it the mathematical contents of a neural representation

and where they propose a uh based upon eagan 2019

a computational theory proper for representation

and so the aspects of representation that they're going to pursue

are the mathematical function that's being realized

that's like the uh the mathematical component then there's the specific algorithms or the process theory

by which the system actually does

compute that so it's one thing to say

well the ants are doing a you know np hard

optimization problem it's another thing
to ask what are the rules that each
ant is engaged in such that that
optimization
is the outcome third are the questions
about representational structures
so that's like not what is the ant
colony doing or what does one ant
do but what is the representation of the
ants across the whole colony through
time
as it solves the problem
what are the computational processes
that are defined
over representations so the
representation
is that distribution of ants in the
colony the distribution of active neural
states in the brain
and what is the computational process
that is defined
over representations that links them
together so that they accomplish
the mathematical function and then
lastly is this ecological component
which is really um
so critical which is the recognition
that the
internal dynamics of the system one
through four
the math the algorithm the
representation the computational
component
that's introspective and we're never
going to understand
function or adaptiveness or
functionality without
also understanding ecologically what is

the context that the algorithm is
occurring within
um and then is there any thoughts on
that from either of you two
cool i thought this was really a nice
way to
come from a really well-grounded uh
definition of what all these different
components of representation
and then in the end ground it to the
real system which is the ecosystem
okay um now they're going to add this
heuristic
cognitive content and here we can look
at
figure four so in figure four
this is the deflationary account of the
contents of representation so remember
deflationary just means it's just a
party
okay it's not a big deal it's just a
party
so here is how they're gonna go about
deflating
representations and they're gonna split
it into
two parts that aids in the deflation
it's kind of like you know when you get
a package and there's the air bubbles
and it's partitioned you need to deflate
each one of those partitions
so they're gonna separate the world into
two kinds of components
cognitive and computational and then
they're gonna deflate
um those two aspects so computationally
we're going to draw off of those eagan
uh

terms that we just went through one
through five so there we have
mathematical
functions algorithms representations
computational processes and ecology
those are all computational like in the
sense of the ecological component
whatever it is it's just that there's no
magic in the ecology
there's no magic in the computation and
then
over on the left side here the cognitive
component
it's actually there's just one air
bubble to deflate cognition is just
about intention
the intentional gloss is what they um
call it here
here the cognitive content is taken uh
oh wait for that little thing to
disappear
uh as an it is taken in an anti-realist
sense
in a type of explanatory gloss that only
has an explanatory
instrumentalist or remember fiction
organismal centered fiction
role as the interpretation given to the
neural representation
by the experimenter like oh when this
set of neurons is activated the mouse
wants to drink and so you could say
computationally
here's the mathematics here's the
algorithm that's happening the
representation the computation and the
ecology with the water
those are all just what they are and

then there's the cognitive element which
we've now
isolated so that's really where we've
gotten to is we can isolate the
cognitive part and say
this is the intentional gloss this is
the mouse
thinking about being thirsty
experiencing being thirsty wanting to be
thirsty wanting water
so that's a really nice partitioning
and that's still within the traditional
frameworks of computational neuro as
well as cognitive science
any thoughts on figure four
cool um
then we go back to the paper we hear a
little bit more about
fictionalism in the philosophy of
science
and suffice to say there's a long uh and
storied history
of fictionalism in science uh
because science is always making models
that are as if they're true about the
world
but here's where they say
they're used by scientists to explain
intentional behavior they're models used
by scientists
again this is kind of a circular
definition it's like saying
cognitive neuroscience is whatever
cognitive neuroscientists do
it's like yes it's true um and then we
can also look beyond that
and they discuss this this idea of
empirical adequacy

or true enough and that's a key idea
and uh it's kind of like when people say
let the data speak
or uh the data are clear well the data
were clear
on epicycles and people were making more
and more epicycles and getting more and
more empirical adequacy
and it was true enough for the people
who were using them but it was wrong
so empirical adequacy always must be
interpreted in a really
holistic way where even if the model
precision is extremely high
even if the model precision is
increasing you still could be just
locally overfitting and totally barking
up the wrong tree
um okay finally
near the end of our discussion we reach
this variational semantics
where we're going to actually bring all
these things together
we have the generative models active
inference in the free energy principle
on one side we have semantics and then
we have
phenotype and we're gonna say phenotype
is
just phenotype the brain's just doing
brain stuff the arm is just an
arm a frog is just a frog and semantics
are just that they're just
meaning but how are we going to bridge
these two worlds
how are we going to go from just neural
representation
to just meaning

well first they bring up the viewpoint
that the deflationary view of
representation
downplays the role of the fifth
ecological component of the
computational theory proper
so what their claim is here is that um
deflationary
representationalism from a computational
perspective which is like mainstream
science
discusses the first four points here a
lot
but they ignore the ecological component
so i certainly resonate with this
critique
and that's why earlier i said this is
one of the strengths of the free energy
principle is it reminds us
even when we're drilling down into the
thirst center of the mouse
how could you ever really extract that
from the ecological context
where there's water or no water um the
body states
where it's uh dehydrated or not and how
could you even extract that from the
evolutionary history of water being
available or being associated with other
sensory cues
um here's where they connect it to the
free energy principle
they're going to say the formalism that
is underwriting the free energy
principle
licenses or allows us to have a crucial
observation
which is that the mathematical

structures and processes
are defined over a state space remember
information geometry
and implicitly over an associated belief
space or a statistical manifold
so manifold is a lower dimensional
representation of a higher dimensional
uh
pattern so if you have some sort of high
dimensional pattern but you project it
into low dimensions
it might exist on a line in two
dimensions even though it was a cluster
in 100 dimensions
and that's sort of the idea of these
manifold techniques and
another mapping that to bring it one
level closer to the paper
is like there's a manifold that's a
continuum thirsty to not thirsty
and then there's so many neural states i
mean we have billions of neurons and
other cells
maybe individual neural states are never
replicated ever because each one is so
unique and the brain is always changing
but you can project down onto this
internal manifold which is thirsty to
not thirsty
across your life even as the number of
neurons you have changes or the neurons
fundamentally change
themselves um so we're going to use
this idea that the environment is
important and that the internal states
are mapping onto an internal statistical
manifold
um specifically an action-oriented one

and then

we're gonna bring that to semantics any

thoughts before we move forward

okay cool so

it is often noted that one does not

obtain semantic content from mere

systematic covariation

so that's sort of the uh helen keller

initial

learning approach which was like she

learned famously

by having water on her hand and then

having sign language

spell out water on her hand so that was

an instantaneous co-variation

between a symbol string of sign language

and a sensory experience of water

and so that was like the rosetta stone

for linking

an external event and sensory sequence

to

um symbols that that's uh lexical

semantics delivered as sign language but

that's not how we get all of our

semantic content

um under the free energy principle there

is an implicit semantics at play

that is baked into the system's dynamics

by evolution

so there's something about sugar that

tastes good there's something about

warmth at the right level that feels

good um

that is something that evolution has

endowed us with because we're not simply

randomly assembled combinations of

phenotypes we

we've been ecologically selected to

experience

a semantic quality to different types of
sensory experiences

and that's not to say that they're not

plastic or they can't be learned

or nuanced can't be added but there is

uh potentially within this trajectory of

states on the internal manifold

there actually is the hope of a formal

semantics that could fall out of this

system's dynamics

and therefore be characterized

mathematically but the hope for a formal

semantics

that arises from phenotype

representation is there

because the world actually does have

order and stimuli actually do

have valence in the world any thoughts

on that

okay here's where we take stock in

section 4.2

we have we have retained the general

description of representational content

from the deflationary account

so in other words we got what we wanted

out of the deflationary account which is

to say

brains are just brains ecosystems are

just ecosystems

meaning is just meaning we deflated

everything else away

now we're left with just what's there we

now

will use this deflationary model to

specify a computational theory proper

that leads to a formal semantics via the

free energy principle

so let's return here here's the fallacy

um

that people were making they would

define mathematics

algorithms representation computational

processes of a system

bacteria or the mouse or a simulation

but

it was always focused on the system of

interest

and so without specifying the ecology it

was kind of like building a castle on

nothing

because the internal representations the

computational components

were not grounded in the evolutionary or

ecological niche

of function and therefore the cognitive

component and the intentional gloss was

kind of open-ended

it's like okay this neuron is activated

when the mouse is drinking water

so maybe that means the mouse is thirsty

um

the missing piece was the ecological

component

um then they're going to they say

figure four i believe that they mean

figure 5

uh or maybe they are referring to the

previous figure 4.

but here is where they're going to

reformat egan's

uh scheme which is here on the screen

they're going to reformat that in terms

of the paradigm of free energy principle

so this is like the big um synthesis

piece

this is you know almost at the end of
our discussion where we've really
reached
their synthesis they're saying look the
common framework is computational
dynamics
not computational components and
cognitive components
that are you're going to split up but
it's all about computational
dynamics that's as deflationary as you
can get
we separated and deflated into a
cognitive component
and then guess what we realize that by
having an ecological component
we don't even need intention because
intention oh well maybe if this neuron
is activated the mouse is thirsty
now you don't need to worry about that
you can just say
the mouse is this ecological agent that
has to reduce its uncertainty about how
thirsty it is
and using that partitioning of the
fisher information equations that we
kind of passed
over earlier we can formalize the
ecological relationship
of the mouse to the surroundings and by
doing so
we can get intentional like claims like
the mouse wants to drink water it's
trying to open up the water bottle
it's running as fast as it can to get
water we get these types of intentional
like claims coming out of here from the
ecological component

vis-a-vis dual information geometry and
so here is where they unpack
each of these components of eagan within
the free energy principle
so the mathematical function that is
being enacted
is the free energy functional which is
the minimization of expected
free energy given the generative model
of the organism
the algorithm that the organism uses to
carry this out
is stochastic variational gradient
descent on free energy
um which will show one little thing
after this the representational
structure uh are the internal states of
the model
reflected by these lower dimensional
manifolds
and then the computational process by
which these representations
morph and evolve is active inference and
then
we're grounding it in the piece that is
most lacking for mainstream science
which is the ecological component
specifically formalized here as dual
information geometry
okay that's the big
claim any thoughts on that or how does
that strike both of you
after like hearing that whole lead-up
i think this figure sums it up really
nicely and draws the kind of
parallels that encompass
the setup of the aims so it's really
nice and i agree that the ecological

component
is is lacking and
is quite important in actually
understanding
what the system is doing
cool yvonne anything
okay so let's return to the paper
um much hangs philosophically
on what it means to represent a target
domain in terms of the relationship
between
mental states mind and physical states
that realize them
brain so now we're talking about the
mind brain
mapping mind is the intentional gloss
that's like well the mind wants this and
then the question
has always been well how is the mind
related to the brain
and we're not coming at this from the
question of how is consciousness
generated in the brain or other
uh areas that also deal with mind and
brain duality
but just specifically using
representationism as the wedge
there's a representation of wanting
water and then there's the brain state
and that's what we've been all about is
exploring this connection
between wanting water and actual neural
firing patterns
and non-neural things happening like
hormone levels and all these other
components
that aren't even being attempted to be
described by

neural network models or other features
um there now this is where they
philosophically tie it all
together and again this fristen at all
2020

paper uh there's a lot more philosophy
to delve into and the citations and this
these will be papers we explore in
future discussions
but um of the philosophical perspectives
that relate mental and physical states
mind and brain

ours is most consonant with
functionalism and therefore the multiple
realizations that it entails

functionalism is the view that features
that characterize mental states
are not intrinsic features of a state
but rather the function which is the
input

output mapping between that state and
other states of the system

so here's where they're tying it to a
lot of other previous
ideas related to cognitive science and
other areas

then they returned to some of those
earlier questions like

can you get a misrepresentation and they
then talk about some future research

where you actually are able to
objectively ask how well

does the sensory data conform to
hypotheses about what caused them

so that would open up the door to
misrepresentation not that it's a binary
state correct or incorrect

representation but you'd be able to look

at two different representations
and ask which one was more accurate you
know if you have two maps of the subway
and one of them is causing you no
surprises when you're
using it and the other one there's
outdated stops and the schedule is off
it's pretty clear that that state
representation
of the map is less adequate it's less
good of a model
and that gives us this key feature of
representations being able to
misrepresent
which is important for being able to act
well but we're also not going to fall
into this absolutist trap
where there's like a best representation
or only representation
um and that's actually the last
paragraph they basically ask
that yes under the free energy principle
there are structures internal to an
organism that are the bearers of
semantic content
we can specify these internal structures
in terms of
deflationary computational theories
figure five
um and by virtue of the double
information geometries that are in play
under the free energy principle that's
the part the thermodynamic or the
thermo-informational partitioning of
internal and external states by markov
blankets
so we'd make this dual information
geometry ecological partitioning

and then we actually can use our
mathematical account
to reach an implicit semantics which is
the set of hypotheses about underlying
causal factors
that the system is parsing in order to
make sense of its sensory stream
this might be seen as vindicating the
structural representationalist account
um in other words so if it's
deflationary are we
uh vindicating the idea that it's just
neurons
but there's a critical twist the
critical twist
is that those structures that bear
contents are not merely
neural representations but indeed
phenotypic representations
versus the internal states of an
organism given the markovian partition
that bear this content and this is like
a really
um amazing way to close the paper
because it's saying
yes neurons are involved but depending
on what level the cognitive process is
playing out on
that is going to co-define where the
markovian blanket
is so for example if multiple people are
on a team together
and the team is engaged in collective
cognition then
that team markov blanket is actually the
one
that is bearing the representational
content so multi-party

dialogue is actually being represented
at the group level
though there also could be smaller
markov blankets that could be drawn
around each person each brain region
each neuron each organelle
everything is always fractal fractal
hierarchical embedded
but for utility we can actually draw out
these uh
partitions as they've suggested so that
is the end of the paper
it's almost the end of our time and i
think it's just good if we
um had any closing thoughts before we
shut it down but this was really an
awesome discussion
so maybe what's one thing that you took
away from the paper or one thing that
you're still wondering about at the end
of it
i just i just understand i need to read
it again
one more time i had i had your
presentation and
i go through the through the paper with
it but
still i need to dip more
perfect always we can dig deeper sasha
um yeah that was really useful to walk
through
this whole paper um incredibly dense but
i like what you said about the last
closing statement is that it's kind of a
um yeah free energy is the lens to look
through
and by placing the markov blanket at
different

places in the system it can be
scientifically useful
for us to understand the system
as it's useful to the organism that's uh
using that representation
well said well said so
that concludes our discussion thanks a
lot for
listening everybody who's listening live
or in replay
this has been a team com podcast number
two
and um we are always open to
participants
and to suggestions about how we could
discuss these ideas
better how we could bring it to the
people in a more exciting and
participatory way
so get in touch with us thanks again for
listening and i'm going to